Geometric Digital Twin Verification Tool
User Guide

What Is It?

The Geometric Digital Twin (GDT) Verification Tool is a prototype web application designed to compare two versions of a GDT:

- Model G(0): the reference geometric model
- Model G(t): the newly generated model

The tool applies to photogrammetry-based models and evaluates whether the new model meets three criteria—accuracy, level of detail (LoD), and resolution—against the reference. If all criteria are satisfied, the new model is considered suitable for updating.

App Overview: Three Main Tabs

1. Model G(0) – Input reference model data
2. Model G(t) – Input newly generated model data
3. Verification Results

Tab 1: Model G(0)

Step 1: Geometric Digital Twin Characteristics

- Select the scale (e.g., Building Part, Building, Site, Urban) - *optional*
- Scale Value (enter a specific value for the scale) - *optional*
- Select Building Life Cycle Stage (Pre-construction, Use, End of Life) - *optional*
- **Enter Average Ground Resolution of Model (mm/px)** – *mandatory*

Step 2: Sample for Evaluation

- **Select Sample Type (Feature-based or Scale-based)** - *mandatory*
  - If Feature-based is selected, enter the total number of features/items - *mandatory*
  - If Scale-based is selected, enter the sample scale - *optional*

Step 3: Parameters for Updating

- Number of Previous Updates (enter how many times the model has been updated) - *optional*
- Update Schedule (select Planned or Event-driven) - *optional*
- Survey Type (choose Image-based or Range-based) - *optional*
- Data Acquisition Sequence (select Parallel, Sequential, or Mixed) - *optional*

Step 4: Level of Detail (LoD) Verification

- **The tool suggests a LoD based on your AGR input**
- Verified LoD (verify value for LoD 2.1 – 3.2) - *optional*
- Feature's RMSE (enter the Root Mean Square Error for specific features (e.g., roof elements, windows)) – *optional*

Step 4.1: Ground Sample Distance (GSD) Calculation

- **Enter: Sensor size (mm), Focal length (mm), Flight height (m), Image width (px)** - *mandatory*
- The tool calculates GSD and model resolution

Step 5: Data Quality Elements

- Depending on your sample type, select relevant Data Quality Elements from a checklist
- For each selected measure, provide the required value (number, yes/no, or text)

Download & Save

- Download Model Data (download your input as a CSV file)
- Save & Continue (save your progress and move to the Model G(t) tab)

Tab 2: Model G(t)

- Enter updated values for the new model version (only Step 4.1 is mandatory)
- Fill in values for Data Quality Elements (only those chosen in Model G(0) are available)
- Download Model Data (download Model G(t) data as CSV)
- Save & View Results (save your progress and move to the Verification Results tab)

Tab 3: Verification Results

- Scale & Life Cycle Stage (review the selected values for both models)
- Sample for Verification (review sample type and details)
- LoD Verification Summary (review AGR of models, suggested LoD, calculated resolution, and Feature's RMSE for both models)
- Data Quality Elements Summary (review all selected Data Quality Elements for models)
- Decision Model based on Mandatory Data Quality Elements (enter geometric deviation statistics for model alignment) – *optional, only applies if models were previously aligned in external software (e.g., CloudCompare)*
- **Condition 1:** Model Accuracy Verification (the tool calculates and compares accuracy metrics (D_{acc} *vs* δ_D))
- **Condition 2:** LoD Verification (the tool compares LoD between models)
- **Condition 3:** Model Resolution Verification (the tool compares resolution between models)
- Performance Comparison (shows the percentage of Data Quality Elements where Model G(t) outperforms Model G(0)) – *optional, only if Conditional or Optional DQ Elements were selected*
- Verification Score (a color-coded summary indicating the suitability of Model G(t) for updating)
- Download Results (download verification results as CSV or JSON)

Troubleshooting

- Deployment: The prototype app is currently hosted on Streamlit. If the app does not display properly, try refreshing your browser
- Required Fields: Certain calculations require all relevant fields to be filled. If a result shows "Not calculated," refer to the provided hints ?
- Data Quality Elements: Only the measures selected in Model G(0) are applied to Model G(t) to ensure a fair comparison

Contact: For help or to report issues, contact *iryna.osadcha@ktu.lt*